Shane Holmes

Computer Engineer
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github.com/ShaneHolmes10

Summary

Computer Engineer with experience in embedded systems, robotics, and software development. Passionate about working on novel challenges in an interdisciplinary environment. Affiliations with the Fulbright Killam Fellowship and IEEE HKN.

Education

Northeastern University, Master of Science in Computer Science – Boston, MA — Jan 2024 – Jan 2026

Florida Polytechnic University, Bachelor of Science in Computer Engineering – Lakeland, FL — Jan 2019 – Jan 2023

- Graduated Magna Cum Laude

University of Toronto, Exchange Student in Computer Science – Toronto, ON — Sept 2022 – Jan 2023

- Academic Exchange with Fulbright

Experience

Computer / Robotics / Embedded Engineer, Overhead Intelligence – Lake Wales, FL — Oct 2023 – May 2024

- Worked on the electrical and computer systems of a battery-powered UAV
- Designed and implemented LiDAR-based mapping
- Collaborated with a multidisciplinary team of engineers and participated in in-field deployment
- Technologies: Odroid, STM32, Raspberry Pi, C++/C, Python

Computer / Robotics / Software Engineer, SS Innovations (AVRA Medical Robotics) – Orlando, FL — May 2022 – Sept 2022

- Worked on the computer vision sensor system for a surgical robot arm to identify facial landmarks
- Achieved accurate simulation test results for surgical motion and object identification
- Technologies: Python, OpenCV, Robot arm

Research Assistant / Robotics & Software Engineer, Advanced Mobility Institute – Lake Wales, FL — Jan 2021 – Jan 2022

- Developed a fleet of small robot vehicles to autonomously navigate and coordinate in an environment
- Built electrical motor systems and an ArUco sensor system to identify vehicles
- Achieved vehicle convoy movement and dynamic obstacle avoidance of objects greater than 3 inches consistently
- Technologies: Python, C++/C, OpenCV, Electronics

Skills

Programming Languages: Python, C++, C, Java

Python Libraries: NumPy, Pandas, Matplotlib, OpenCV, Threading, PyTorch, TensorFlow, Keras, Scikit-learn

Embedded & Hardware: Raspberry Pi, Arduino, ESP32, STM32, Odroid, AVR, Linux, Embedded Linux

Robotics & Autonomy: Computer Vision, LiDAR, IMU, SLAM, GPS, UAV

Software Development: Git, OOP, Software Design, Software Testing, Software Documentation, Embedded Systems

Technical Areas: Machine Learning, Artificial Intelligence, Computer Vision, Digital Signal Processing, Robotics, Hardware, Software

Collaborative Tools: Jira, Confluence, Slack, Microsoft Teams, Zoom, Visual Studio

Projects

Mock Rover — FSI Capstone Project — Sept 2022 – May 2023

built a mock rover for the NASA Space Grant Consortium program through Florida Space Institute.

- Led technical integration of embedded control systems designing a Raspberry Pi-based architecture coordinating four AC motors, stereoscopic camera feed, and wireless command interface
- Achieved ~100ft wireless range, 300lb payload capacity, and 5mph traverse speed
- Technologies: Python, C++, Raspberry Pi, Arduino Uno, motor control systems, OpenCV, 3D printing

Hand Robot — Commissioned Project Nov 2020 – Feb 2021

Designed and fabricated a fully articulated robotic hand with individual finger control and accompanying arm.

- Iteratively designed hand and arm in CAD with custom 3D printed components
- Engineered servo control circuits on Arduino platform for individual finger articulation
- Developed desktop application in C++ for real-time articulation control
- Technologies: Arduino, C++, CAD, 3D printing, servo control systems

3D CPU Software Rasterizer Dec 2025 – present

Built a 3D software rendering engine from first principles in C++ for CPU.

- Full graphics pipeline — model space through screen space — implemented manually using 4x4 matrices
- Triangle rasterization using edge functions and barycentric coordinates
- Scene graph with hierarchical parent-child entity system and automatic world matrix propagation
- Technologies: C++, CMake, Eigen, SFML, CppUnitLite

Latent Explorer Apr 2026 – present

End-to-end tool for training and interactively visualizing the latent space of image datasets in the browser.

- Trained a convolutional decoder on ~202k face images with PCA post-processing to surface principal axes of variation
- Exported model to ONNX and built a pure-JavaScript browser interface running inference with no backend
- Technologies: PyTorch, ONNX Runtime Web, scikit-learn, HDF5

Motor Babbling — RL for Inverse Kinematics Feb 2026 – Apr 2026

Implemented DQN, DDPG, and SAC to teach a simulated planar robot arm to reach arbitrary targets in MuJoCo.

- Compared value-based (DQN), deterministic actor-critic (DDPG), and stochastic actor-critic (SAC) against the same environment; SAC achieved ~80% success
- Built the MuJoCo environment with a configurable number of links and a shared BaseAgent interface for clean algorithm comparison
- Technologies: Python, PyTorch, MuJoCo, Gymnasium, NumPy, Matplotlib

Relevant Coursework

Mathematics: Probability and Statistics, Differential Equations, Linear Algebra, Discrete Mathematics, Analytical Geometry and Calculus (1, 2, 3)

Computer Engineering: Physics (1, 2), Computer Architecture and Organization, Microcomputers, Digital Logic Design, Circuits (1, 2), Digital Electronics, Computer Systems

Software Engineering: Operating Systems Concepts, Data structures and Algorithms, Object Oriented Programming, Computer Programming, Computer Networks, Secure Software engineering, Programming Design Paradigms, Database Systems, Foundations in Software Engineering

Robotics Systems: Autonomous Robotic Systems, Digital Signal Processing, Systems and Signals, Kinematics and control of Robotic Systems, Artificial Intelligence (1, 2), Machine Learning (1, 2), Computer Vision

Awards

Fulbright Canada Killam Fellowship: Fulbright Canada · 2022 · Competitive fellowship awarded to outstanding students for academic exchange between Canada and the United States.

Magna Cum Laude: Florida Polytechnic University · 2023-05 · Graduated with high academic distinction.

IEEE HKN Member: IEEE Eta Kappa Nu · 2023 · Member of the IEEE honor society chapter recognizing excellence in electrical and computer engineering.

President's List: Florida Polytechnic University · 2023

Bright Futures Scholarship: State of Florida · 2019

Florida Polytechnic University Institutional Scholarship: Florida Polytechnic University · 2019

Languages

English: Native speaker